Code No: 136FU **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, August/September - 2024
REMOTE SENSING AND GIS
(Common to EEE, ME, CSE)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	What are the types of aerial photographs? And explain any one in detail.	[2]
b)	Write short note on Differential parallax.	[3]
c)	What is the importance of Resolution?	[2]
d)	Draw Electromagnetic Spectrum.	[3]
e)	What is a Projection?	[2]
f)	Explain the components of GIS.	[3]
g)	What are Common Topological Errors?	[2]
h)	How to define Coverage in Vector data format?	[3]
i)	What is a Metadata?	[2]
j)	Explain about IMGRID model.	[3]

PART – B

(50 Marks)

2.a)	Present a detailed classification of sensing devices.	
b)	Explain the Geometric properties of aerial photographs.	[5+5]
	OR	
3.a)	Explain the Parallax Measurement using Fiducial Line.	
b)	Explain the measurement of Height based on Relief displacement.	[5+5]
4.a)	Explain the components of remote sensing with a neat sketch.	
b)	Explain False Colour Composite significance in the field of RS.	[5+5]
	OR	
5.a)	Explain the advantages of Remote Sensing.	
b)	Discuss the essential features of visual interpretation techniques.	[5+5]
6.a)	Explain the methodology of linking spatial data with attribute data.	
b)	Give the classification of map projections with examples.	[5+5]
	OR	
7.a)	Explain the role and importance of data structure in GIS.	
b)	What are the GIS layers developed for watershed characterization? Explain.	[5+5]

8.a) Give notes on Geometric representation of Spatial features in GIS.
b) List out various spatial rules that define a Topology. [5+5]

OR

9.a) Explain about TIGER and Topological models in Vector GIS.
b) Explain the process of DTM Generation. [5+5]

10.a) What are the advantages and disadvantages of the Raster Data Models?
b) What are the three raster GIS models? Give advantages of each. [5+5]

OR

11.a) How you will identify and rectify the various types of errors during digitization?
b) Explain Various Methods used for digitization to create spatial data. [5+5]

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